



In this workspace, you'll scrape the novel Moby Dick from the website [Project Gutenberg](#) (which contains a large corpus of books) using the Python `requests` package. You'll extract words from this web data using `BeautifulSoup` before analyzing the distribution of words using the Natural Language Toolkit (`nltk`) and `Counter`.

The Data Science pipeline you'll build in this workspace can be used to visualize the word frequency distributions of any novel you can find on Project Gutenberg.

```

# Import and download packages
import requests
from bs4 import BeautifulSoup
import nltk
from collections import Counter
nltk.download('stopwords')

# Request and encode the text
url =
'https://s3.amazonaws.com/assets.datacamp.com/production/project_147/datasets/2701-
h.htm'
r = requests.get(url)
r.encoding = 'utf-8'

# Extract the text
html = r.text
print(html[0:2000])

# Create a BeautifulSoup object and get the text
html_soup = BeautifulSoup(html, "html.parser")
moby_text = html_soup.get_text()

```

```

[nltk_data] Downloading package stopwords to /home/repl/nltk_data...
[nltk_data] Package stopwords is already up-to-date!

```

```

<?xml version="1.0" encoding="utf-8"?>

<!DOCTYPE html
  PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
  "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd" >

<html xmlns="http://www.w3.org/1999/xhtml" lang="en">
  <head>
    <title>
      Moby Dick; Or the Whale, by Herman Melville
    </title>
    <style type="text/css" xml:space="preserve">

      body { background:#faebd0; color:black; margin-left:15%; margin-right:15%;
text-align:justify }
      P { text-indent: 1em; margin-top: .25em; margin-bottom: .25em; }
      H1,H2,H3,H4,H5,H6 { text-align: center; margin-left: 15%; margin-right: 15%; }
      hr { width: 50%; text-align: center;}
      .foot { margin-left: 20%; margin-right: 20%; text-align: justify; text-indent:
-3em; font-size: 90%; }
      blockquote {font-size: 100%; margin-left: 0%; margin-right: 0%;}

```

```
.mynote {background-color: #DDE; color: #000; padding: .5em; margin-left:
10%; margin-right: 10%; font-family: sans-serif; font-size: 95%;}
.toc { margin-left: 10%; margin-bottom: .75em;}
.toc2 { margin-left: 20%;}
div.fig { display:block; margin:0 auto; text-align:center; }
div.middle { margin-left: 20%; margin-right: 20%; text-align: justify; }
.figleft {float: left; margin-left: 0%; margin-right: 1%;}
```

```
# Tokenize the text
tokenizer = nltk.tokenize.RegexpTokenizer('\\w+')
tokens = tokenizer.tokenize(moby_text)

# Convert words to lowercase
words = [word.lower() for word in tokens]
print(words[:8])

# Load in stop words
stop_words = nltk.corpus.stopwords.words('english')
print(stop_words[:8])
```

```
['moby', 'dick', 'or', 'the', 'whale', 'by', 'herman', 'melville']
['a', 'about', 'above', 'after', 'again', 'against', 'ain', 'all']
```

```
# Remove stop words
words_no_stop = [word for word in words if word not in stop_words]
print(words_no_stop[:5])

# Count the frequency of words
count = Counter(words_no_stop)
top_ten = count.most_common(10)
print(top_ten)
```

```
['moby', 'dick', 'whale', 'herman', 'melville']
[('whale', 1246), ('one', 925), ('like', 647), ('upon', 568), ('man', 527), ('ship',
519), ('ahab', 517), ('ye', 473), ('sea', 455), ('old', 452)]
```